

Temperature And Humidity Monitoring With Email And SMS Alerts

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In this project ESP-12E WeMos D1 Wi-Fi Uno based ESP8266 shield for Arduino-compatible board and Ubidots website platform are used for sending email or SMS alerts for temperature and humidity monitoring applications. The WeMos D1 uploads the temperature data received from a temperature sensor and continuously sends to Ubidots IoT platform. If the temperature exceeds the set value, an email message is instantaneously sent to the field manager.

The authors' prototype of the project showing the WeMos D1 controller board and the DHT11 temperature and humidity sensor is shown in Fig. 1. The interfacing of the sensor with the controller is shown in Fig. 2.

Major components used in this project are described below.

WeMos D1 R1. WeMos D1 is a Wi-Fi development board based on ESP8255 12E. Its functioning is similar to that of NodeMCU, except that the hardware is built resembling Arduino Uno R1. The board looks and works (in most cases) like an Arduino Uno. Apparently, several shields, sensors, and output devices that are manufactured for the Arduino platform will work on WeMos D1 with the added advantage of built-in Wi-Fi. The WeMos D1 board comes without a USB cable. DHT11 sensor is connected to pin D2 of WeMos D1. The WeMos D1 R1 board used here has one analogue input (3.2V maximum input voltage) and eleven digital input/output pins. (You can also use WeMos D1 R2 board but note the change in position of pin D2.)

DHT11. It is a temperature cum humidity sensor that has four pins. Pins 1, 2, 3, and 4 correspond to Vcc, Vout (output), no connection

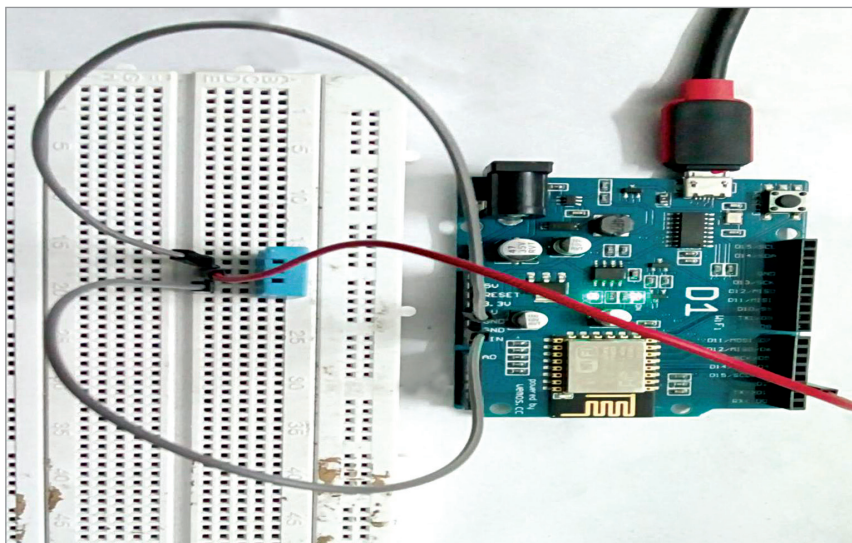


Fig. 1: Authors' prototype

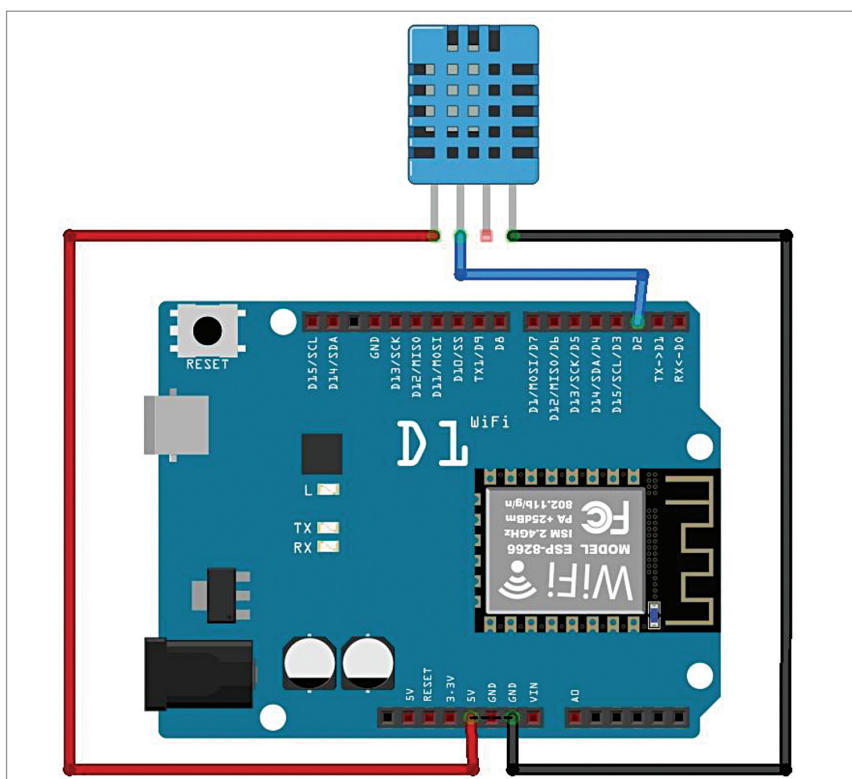


Fig. 2: Interfacing of DHT11 sensor with WeMos D1 board